Walks, paths, and connectivity from the adjacency matrix: a short account

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Abstract

For a finite simple undirected graph with adjacency matrix A, the entry $(A^k)_{ij}$ is the number of walks of length k from vertex n_i to vertex n_j (walks, not paths). A path is a walk with no repeated vertex. A graph on N vertices is connected if and only if every off-diagonal entry of $A + A^2 + \cdots + A^{N-1}$ is positive. Each statement is proved and the test is run on two graphs.

1 Introduction

Let G be a finite simple undirected graph with N vertices $n_1, \ldots, n_N$ and adjacency matrix A (defined in section 2). The lecture board asks two questions about it, and the assignment adds a third [2, 3, 7].

(Q1) What does the entry $(A^k)_{ij}$ represent? The mark above A on the board is read as an exponent k [7]. *Answer:* it is the number of walks of length k from n_i to n_j (theorem 3.1); it counts walks, not paths.

(Q2) How can one tell from A whether G is connected? *Answer:* G is connected if and only if every entry off the diagonal of $B = A + A^2 + \cdots + A^{N-1}$ is positive (theorem 5.3).

(Q3) What is a path between n_i and n_j? *Answer:* a walk from n_i to n_j in which no vertex occurs twice (definition 4.1).

The answers come in the order Q1, Q3, Q2, because the test for Q2 needs the path. A full-text search of the course textbook [6] finds neither the walk-counting theorem nor a matrix test for connectivity, so both are taken from other sources, and every proof is written out in full; "derived" marks this paper's own proof of a result stated in a cited source. Textbook page numbers are printed ones, one lower than the PDF's.

2 Setting and definitions

Setting. $G = (V, E)$ is a graph in the sense of the textbook: a set $V = \{n_1, \ldots, n_N\}$ of vertices, $N \geq 1$, and a set E of edges [6, Definition 1.1.1, p. 9]. It is finite, simple (no edge from a vertex to itself, at most one edge between two vertices) and undirected (the edge joining u and v, written uv, is the same as vu) [7]. The vertices u and v are *adjacent* if $uv \in E$ [6, Definition 1.1.4, p. 11]. The indices i, j, r run over $1, \ldots, N$; $k \geq 0$ is an integer.

Definition 2.1 (Adjacency matrix)**.** The *adjacency matrix* of G is the $N \times N$ matrix A with $A_{ij} = 1$ if n_i and n_j are adjacent and $A_{ij} = 0$ otherwise [6, p. 11]; the board states the same rule in other letters [2, 3].

Its entries are 0 or 1. Matrix powers are the usual ones: $(XY)_{ij} = \sum_{r=1}^{N} X_{ir}Y_{rj}$, $A^0 = I$ (the identity matrix) and $A^{k+1} = A^k A$. Every entry of A^k is a non-negative integer, since sums and products of non-negative integers are non-negative integers (*used in* theorem 5.3).

Definition 2.2 (Walk)**.** A *walk* from u_0 to u_k is a sequence of vertices $u_0u_1 \cdots u_k$ in which each vertex is adjacent to the one before it; its *length* is k, the number of edges used, each counted as often as it is used. Vertices and edges may repeat. This is the textbook's Definition 1.2.10 [6, p. 21], which lists edges as well; in a simple graph the vertices fix the edges. This paper also allows $k = 0$: a single vertex is a walk of length 0 from itself to itself. This is the paper's own convention; the textbook's usage assumes it [6, pp. 23, 49].

Definition 2.3 (Path)**.** A *path* is a walk whose vertices are pairwise distinct. This is the course convention [7]. It is the textbook's *simple path* [6, Definition 1.2.13, p. 22]; the textbook's own word "path" means a walk with no repeated edge, and this paper never uses the word in that sense.

Definition 2.4 (Connected)**.** G is *connected* if for any two vertices u, v there is a path from u to v. The board states this [2, 3]; the textbook's Definition 2.1.1 [6, p. 48] is the same with the textbook's own "path". A vertex is joined to itself by the path of length 0, so only pairs of different vertices matter.

Board graph. The board's matrix has rows 0 1 0, 1 0 1, 0 1 0 [2, 3]:

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}.$$

It encodes the graph with edges n_1n_2 and n_2n_3, which is the example graph throughout. The board's drawing shows a third line that the matrix does not encode; the matrix defines the example.

3 Answer to Q1: the entries of A^k count walks

Theorem 3.1 (Walk counting)**.** *For every integer $k \geq 0$ and all i, j, the entry $(A^k)_{ij}$ is the number of walks of length k from n_i to n_j.*

The University of Galway notes state this for $k \geq 1$ with an induction proof [4, Theorem 1.1.3, p. 4]; the case $k = 0$ is this paper's convention. The proof is derived here.

Proof. By induction on k.

Base case $k = 0$. $A^0 = I$. A walk of length 0 from n_i to n_j is a single vertex that is both start and end, so there is exactly one such walk if $i = j$ and none if $i \neq j$. This is I_{ij}.

Inductive case. Assume that for some $k \geq 0$ and all i, r the entry $(A^k)_{ir}$ is the number of walks of length k from n_i to n_r. Fix i and j. Since $A^{k+1} = A^k A$,

$$(A^{k+1})_{ij} = \sum_{r=1}^{N} (A^k)_{ir} A_{rj}. \tag{1}$$

A walk $u_0 \cdots u_k u_{k+1}$ of length $k+1$ from n_i to n_j has $u_k = n_r$ for exactly one r; sort the walks by this r. Fix r. If n_r is adjacent to n_j (so $A_{rj} = 1$), the walks of this kind correspond one to one to the walks of length k from n_i to n_r: deleting the final vertex n_j gives such a walk, and appending n_j to such a walk gives a walk of the kind, because n_r is adjacent to n_j. By the hypothesis there are $(A^k)_{ir} = (A^k)_{ir}A_{rj}$ of them. If n_r is not adjacent to n_j (so $A_{rj} = 0$), no walk can end with the step from n_r to n_j, so there are $0 = (A^k)_{ir}A_{rj}$ of them. Every walk is counted for exactly one r, so the number of walks of length $k+1$ from n_i to n_j is $\sum_r (A^k)_{ir}A_{rj}$, which is $(A^{k+1})_{ij}$ by eq. (1). □

Warning: walks, not paths. The proof uses only that consecutive vertices are adjacent, so repeated vertices are counted. For a path, appending n_j may repeat a vertex and the one to one correspondence fails. In the board graph $(A^2)_{11} = 1$ and the walk counted is $n_1n_2n_1$, which repeats n_1; the only path from n_1 to n_1 is the single vertex n_1, of length 0 (a path of length $k \geq 1$ from a vertex to itself would repeat it). So "$(A^2)_{11}$ counts the paths of length 2 from n_1 to n_1" is false, although "counts the walks" is true. Also $(A^4)_{22} = 4$ by script, so the entry is not "1 if n_j can be reached from n_i".

4 Answer to Q3: the path between n_i and n_j

Definition 4.1 (Path between n_i and n_j)**.** *A path between n_i and n_j* (also a path *from n_i to n_j*) is a sequence $u_0, u_1, \ldots, u_k$ with $k \geq 0$, $u_0 = n_i$, $u_k = n_j$, in which u_{t-1} and u_t are adjacent for $1 \leq t \leq k$ and the vertices $u_0, \ldots, u_k$ are pairwise distinct. Its length is k.

This is definition 2.3 with the ends fixed, the textbook's Definition 1.2.13 with ends n_i and n_j [6, p. 22]. If $i \neq j$ such a path has at least one edge; if $i = j$ the only path is (n_i), since $u_0 = u_k$ with $k \geq 1$ would repeat a vertex. In the board graph $n_1n_2n_3$ is the only path from n_1 to n_3.

5 Answer to Q2: a connectivity test from A

The test rests on two facts about walks and paths. Both are stated in Grinberg [1, Corollary 2.9.7, p. 39, Section 2.9.4, p. 40] and the first is the textbook's Lemma 1.2.1 for $u \neq v$ without a length bound [6, p. 23]; the proofs are derived here.

Lemma 5.1 (Every walk contains a path)**.** *Let $W = u_0 \cdots u_k$ be a walk from u to v. Then there is a path from u to v of length at most k.*

Proof. By strong induction on k: for walks of length k we assume the statement for all shorter walks. *Base case* $k = 0$. $W = (u)$ has one vertex, so it is a path of length 0. *Inductive case* $k \geq 1$. If the vertices of W are pairwise distinct, W is a path of length k. Otherwise there are positions $s < t$ with $u_s = u_t$. Let $W' = u_0 \cdots u_s u_{t+1} \cdots u_k$ (if $t = k$, $W' = u_0 \cdots u_s$). It is a walk, because the only new consecutive pair is u_s, u_{t+1} and $u_s = u_t$ is adjacent to u_{t+1}. It runs from u to v and has length $k - (t - s) < k$. By the hypothesis W', and so W, yields a path from u to v of length at most $k - (t - s) \leq k$. □

Lemma 5.2 (Reachability from the powers)**.** *Let $i \neq j$. There is a path from n_i to n_j if and only if $(A^k)_{ij} > 0$ for some k with $1 \leq k \leq N - 1$.*

Proof. ($\Rightarrow$) Let a path from n_i to n_j have length k. As $i \neq j$, $k \geq 1$. A path of length k has $k + 1$ pairwise distinct vertices among the N vertices of the graph, so $k \leq N - 1$. A path is a walk, so by theorem 3.1 $(A^k)_{ij} \geq 1$. ($\Leftarrow$) If $(A^k)_{ij} > 0$ with $1 \leq k \leq N - 1$, then by theorem 3.1 a walk of length k joins n_i to n_j, and by lemma 5.1 a path joins them. □

Theorem 5.3 (Sum test)**.** *Let $N \geq 2$ (for $N = 1$ the graph is connected) and $B = A + A^2 + \cdots + A^{N-1}$. Then G is connected if and only if $B_{ij} > 0$ for all $i \neq j$.*

This is Theorem 2 of Paramadevan and Sotheeswaran [5, p. 17]; the proof here is derived.

Proof. Each A^k has non-negative entries, so $B_{ij} = \sum_{k=1}^{N-1} (A^k)_{ij}$ is a sum of non-negative terms, and it is positive exactly when at least one term is positive. For $i \neq j$ this holds exactly when a path joins n_i and n_j, by lemma 5.2. So $B_{ij} > 0$ for all $i \neq j$ exactly when every pair of different vertices is joined by a path, which is definition 2.4. □

The cutoff $N - 1$ is the largest length a path can have (proof of lemma 5.2). The entries of B are walk counts, not path counts; only whether they are zero is used.

6 Worked example and limits

All matrices and walk lists below were computed by the scripts in the `scripts/` folder, which also holds their outputs.

Connected: the board graph ($N = 3$, edges n_1n_2, n_2n_3). Here $B = A + A^2$ with

$$A^2 = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 1 \end{pmatrix}.$$

By theorem 3.1, $(A^2)_{11} = 1$ counts the walk $n_1n_2n_1$, $(A^2)_{13} = 1$ counts $n_1n_2n_3$, $(A^2)_{22} = 2$ counts $n_2n_1n_2$ and $n_2n_3n_2$, and $(A^2)_{12} = 0$ because no walk of length 2 joins n_1 to n_2. The script also gives $(A^3)_{12} = 2$, for the walks $n_1n_2n_1n_2$ and $n_1n_2n_3n_2$, and $(A^4)_{22} = 4$. All off-diagonal entries of B are positive, so G is connected by theorem 5.3: the pairs n_1n_2 and n_2n_3 get a positive entry from A, and the pair n_1n_3 gets $B_{13} = 1$ from A^2, which counts the path $n_1n_2n_3$.

Disconnected: the edge n_1n_2 and the isolated vertex n_3 ($N = 3$). Then

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad A^2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

The entries $B_{13} = B_{23} = B_{31} = B_{32} = 0$, so G is not connected by theorem 5.3. This agrees with the graph: no walk joins n_3 to n_1 or n_2.

Limits. The test answers the question using A alone, but it has limits. It reads only whether entries are zero: a positive entry says a walk of exactly length k exists, hence a path of length at most k, not a path of length exactly k. The diagonal of B is ignored, and a connected graph can have zeros there: for $N = 2$ and one edge, $B = A$ has a zero diagonal. The sum must run to A^{N-1}: on the path graph with edges n_1n_2, n_2n_3, n_3n_4 ($N = 4$) the script gives $(A^1)_{14} = (A^2)_{14} = 0$ and $(A^3)_{14} = 1$, so a sum stopped at A^2 would call this connected graph disconnected. The test is proved only for finite simple undirected graphs.

References

[1] Darij Grinberg. *An introduction to graph theory (Text for Math 530, Drexel University).* arXiv:2308.04512v3 [math.HO], version of June 7, 2025. 2025. URL: `https://arxiv.org/abs/2308.04512` (visited on Nov. 6, 2026).

[2] *Lecture whiteboard photo 1.* Local file `sources/whiteboard-1.jpg`.

[3] *Lecture whiteboard photo 2.* Local file `sources/whiteboard-2.jpg`.

[4] *Linear algebra notes, 1.1 The Adjacency Matrix (University of Galway notes).* University of Galway, lecture notes. The file carries no author name and no date. URL: `https://maths.nuigalway.ie/~rquinlan/linearalgebra/section1-1.pdf` (visited on Nov. 6, 2026).

[5] P. Paramadevan and S. Sotheeswaran. "Properties of Adjacency Matrix of a Graph and It's Construction". In: *JSc EUSL* 12.1 (2021). Eastern University, Sri Lanka, pp. 13–21. URL: `https://www.fsc.esn.ac.lk/jsc/archive/12.1.2.pdf` (visited on Nov. 6, 2026).

[6] P. A. Petrosyan, V. V. Mkrtchyan, and R. R. Kamalyan. *Graph Theory, study guide.* Local file `Grafneri_tesutyun.pdf`, in Armenian. 2015.

[7] *Summary of user statements.* Local file `sources/user-statements.md`. 2026.